

Unifying spatially-explicit Rao's entropy, spatial concentration, and accumulation curves: the different aspects of local biodiversity

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Abstract

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1. Introduction

2. Methods

2.1 Ripley's K

The framework of spatial point pattern analysis, as introduced by Ripley (1977), relies on the K-function, defined so that $\lambda K(r)$ is the expected number of neighbours located within a distance r of an arbitrary point of the pattern with intensity λ . Its classical unbiased estimator for a point pattern located within a finite sampling window is

$$\hat{K}(r) = \frac{1}{\hat{\lambda}n} \sum_{i=1}^n \sum_{j=1}^n \mathbf{1}(\|x_i - x_j\| \leq r) c(i, j, r) \quad (1)$$

where n is the number of points of the pattern, $\hat{\lambda}$ is the pattern intensity generally estimated as n/A , with A the sampling window area. The indicator function $\mathbf{1}(\|x_i - x_j\| \leq r)$ equals 1 when points i and j are less than r apart, 0 else. $c(i, j, r)$ is correction of edge effects equal to 1 when the circle

of radius r fits into the sampling window. When it does not, the isotropic correction proposed by Ripley (1977) applies to windows of any shape thanks to appropriate geometrical considerations (Goreaud and Pélissier 1999) to estimate the number of neighbors with no bias.

A common extension of this univariate function is the intertype (or bivariate) K12-function proposed by Lotwick and Silverman (1982). Its estimator is:

$$\hat{K}_{1,2}(r) = \frac{1}{\hat{\lambda}_2 n_1} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \mathbf{1}(\|x_i - x_j\| \leq r) c(i, j, r) \quad (2)$$

While $K(r)$ is a multiscale function of second-order neighbourhood properties of a point pattern, which allows robust tests of statistical significance against the null hypothesis of a random distribution of the points within the sampling window, $K_{12}(r)$ allows characterizing the spatial interaction between the two types of points, which can be tested against various null hypotheses, but in particular against the random labelling hypothesis, which means that, conditionally to the point locations, types 1 and 2 are randomly distributed among points of the pattern, or against the population independence hypothesis, meaning that, conditionally to the species-specific spatial patterns, population of type 1 points is independent from population of type 2 points (Goreaud and

Pélissier 2003).

3. Application

3.1 Simulated data

3.2 Paracou

4. Conclusion

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